THE EVOLVING SPECTRUM OF HUMAN AFRICAN TRYPANOSOMIASIS

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ABSTRACT

Human African trypanosomiasis (HAT), or sleeping sickness, continues to be a major threat to human health in 36 countries throughout sub-Saharan Africa with up to 60 million people at risk. Over the last decade there have been several advances in this area, some of which are discussed in this overview. Due to the concerted efforts of several bodies, including better identification and treatment of cases and improved tsetse fly vector control, the number of cases of HAT has declined dramatically. The clinical heterogeneity of HAT has also been increasingly recognised and the disease, while usually fatal if untreated or inadequately treated, does not always have a uniformly fatal outcome. Improved methods of HAT diagnosis have now been developed including Rapid Diagnostic Tests (RDTs). Novel drug treatment of HAT has also been developed, notably NECT for late- stage *T.b.gambiense,* oral fexinidazole for early and the early component of the late-stage of *T.b.gambiense,* and the new oral compounds of the oxaborole group which have shown considerable promise in field trials. Advances in HAT neuropathogenesis have been steady though largely incremental, with a particular focus on the role of the BBB in parasite entry into the Central Nervous System (CNS), and the relevant importance of both innate and adaptive immunity. While the WHO goal of elimination of HAT as a public health problem by 2020 has probably been achieved, it remains to be seen whether the second more ambitious goal of interruption of transmission of HAT by 2030 will be attained.

INTRODUCTION

Human African trypanosomiasis (HAT), also known as sleeping sickness, continues to be a major threat to human health throughout 36 countries in sub-Sharan Africa with about 60 million people at risk from contracting the disease. A neglected tropical disease, HAT is caused by protozoan parasites of the *Trypanosoma* genus which is transmitted by the bite of the tsetse fly of the *Glossina* genus. [1] HAT occurrence is latitudinally determined, occurring between 14^0 North and 29^0 South.[1]There are two clinical forms of the disease, namely West African HAT caused by *Trypanosoma brucei gambiense* (*T.b.gambiense*), and the East African variant caused by *T.b.rhodesiense*. While the former variant accounts for about 95% of HAT cases, and the latter for only about 5% of cases, the clinical picture of these variants is different.[2] The tempo of infection in *rhodesiense* disease is faster than in *gambiense*, with the former disease lasting 6-12 weeks while the latter typically lasts for several months to several years [3,4], probably due to a greater adaptation of the parasite to the host. Untreated, or inadequately treated, HAT is usually fatal, and much current drug therapy for late-stage HAT is toxic.[5] There is a classical distinction between early, haemolymphatic (stage 1) disease in which the trypanosome parasites spread throughout the blood, lymphatic system and peripheral organs, and the late, encephalitic (stage 2) disease in which the blood-brain barrier (BBB) is breached leading to central nervous system (CNS) involvement due to trypanosome invasion.[2] However, while crucial, primarily because of the toxicity of late-stage disease therapies, accurate disease staging is currently very problematic. Over the last 5-10 years there have been several significant advances in our understanding and management of HAT, and the purpose of the current overview is to briefly highlight some of these advances. The latter can be considered under the headings of epidemiological advances, clinical heterogeneity, improved methods of diagnosis, novel therapies, neuropathogenesis, and prospects for the future.

EPIDEMIOLOGICAL ADVANCES

Over the last decade there have been dramatic falls in the numbers of HAT cases detected in sub-Saharan Africa, due to the combined and coordinated efforts of the WHO, African governments, non-governmental agencies, research charities and pharmaceutical companies. The strategy has been to isolate and treat mainly *gambiense* HAT cases thereby limiting disease spread to other individuals, together with improved vector control measures.[1] In 1998 the WHO estimated that there were 300,000 new cases of HAT in Africa.[6] HAT occurrence then declined progressively with these combined measures with just 3796 reported cases /year with <15,000 estimated cases in 2014.[7] A historic low of under 2000 cases were reported in 2017 and under 1000 cases in 2018.[8] Most of the reported cases (mainly *gambiense* disease) over the last five years have been in the Democratic Republic of Congo (DRC) which reported 61% of the cases (mean of 522 cases/year).[8] Most of rhodesiense cases occur in Malawi and Uganda.[7] These figures are very impressive, but certain factors must be borne in mind. First, the history of HAT has been characterised by recurrent epidemics and disease resurgences so careful monitoring of the disease situation needs to be continued long term. Second, it is not possible to exclude both under-detection and under-diagnosis of cases as has been shown to be the case in the DRC.[9] Third, the disease reservoirs for *rhodesiense* HAT are cattle, as opposed to mainly humans in the case of *gambiense* disease, so that form of the disease will be less susceptible than *gambiense* HAT to these successful control measures. It should also be appreciated that in HAT cases in non-endemic countries the disease is more likely to be due to *T.b.rhodesiense.* Thus, it was reported recently that during the decade 2011-2020, 49 cases of HAT were reported in 16 non-endemic countries, 35 of which were caused by *T.b.rhodesiense.* the latter mainly comprised Western tourists who were visiting wildlife regions in Eastern and Southern parts of Africa.[10] These authors reported that the remaining 14 cases were due to *T.b.gambiense* occurring primarily in migrants from Africa who has either originated or visited Western and Central African regions of Africa.

CLINICAL HETEROGENEITY

The clinical features of HAT are classically divided into those associated with the early, stage 1 phase, and the late, stage 2 phase of the disease. There are no definite clinical suspicion criteria that can accurately predict disease staging and the early and late stages often run into each other.[3] Early-stage clinical features typically comprise non-specific symptoms such as fever, lassitude, headache, arthralgia[5] as well as enlargement of peripheral organs such as the liver, spleen, and cardiac involvement, anaemia, eye involvement and endocrine disturbances.[1] Posterior cervical lymphadenopathy typically occurs in *T.b. gambiense* (Winterbottom's sign'). Characteristic late-stage features, occurring after the BBB has been breached allowing parasite entry into the CNS comprise, comprise a characteristic sleep disorder with reversal of the normal sleep/wake cycle, psychiatric disturbances, both pyramidal and extrapyramidal signs, optic neuritis, cerebellar ataxia, sensory disturbances and then finally progressive loss of consciousness, seizures and death.[5,11] Over the last decade, however, it has been recognised that there is a considerable degree of clinical heterogeneity in HAT.

A surprising finding was that some HAT patients in Uganda with proven (see below) early-stage disease had typical neurological features,[12] which questions a distinct difference between the two stages. There are also clear geographical variations in the clinical presentations both within and between countries. In Uganda, clinical features differed in two distinct regions of the country in that in Soroti there was earlier neurological dysfunction and the progression to the late encephalitic stage was faster than in other regions in patients with *rhodesiense* disease.[12] In a different Ugandan region (Tororo), however, patients had more severe neurological disease. Further, in patients with *rhodesiense* HAT, neurological symptoms and signs were more prominent in Tanzania than in Uganda where patients had non-specific features.[13] Also, chronic haemolymphatic features are typical in HAT patients in Malawi.

There are also differences in HAT features in patients in endemic and non-endemic regions. Thus, in tourists and expatriates in the latter areas the disease typically consists of fever, gastro-intestinal disturbances and jaundice though a rapid progression to late-stage disease is typical·[14] In these individuals both sleep disturbances and lymphadenopathy are rare. These features contrast with those seen in HAT patients in endemic countries, It has also become evident over the last decade that in a few patients with *gambiense* HAT is not always fatal if untreated.[15] While, at first, perhaps somewhat counter-intuitive, this has real consequences for disease pathogenesis and management. This is somewhat analogous to the phenomenon of trypanotolerance in certain resistant animals.[1] Such individuals with proven current or past HAT infection are asymptomatic and aparasitaemic, and a few are seropositive. Whether such seropositive individuals should receive anti-parasite therapy, and whether they are a source of infection remain unanswered questions at present.

NEW METHODS OF DIAGNOSIS

A diagnosis of HAT is suspected when typical clinical features occur in an individual within an endemic area, though caution is required to ensure that malaria does not co-exist. In *rhodesiense* disease, the level of parasitaemia is usually high so that microscopic examination of a blood smear or lymph nodes may reveal the presence of the typical trypanosome parasites in which case the diagnosis is established.[2] But it would still be advantageous to have a serological test for *rhodesiense* infection and the ideal qualities of such a potential test have recently been suggested.[16] By contrast, the parasitaemia in *gambiense* disease is cyclical so that parasite detection in the blood is rarer, possibly due to a greater adaptation of the parasite to the host. Serological methods are therefore required for diagnosis in this variant. For many years the Card Agglutination Test for Trypanosomiasis (CATT) was used for this purpose, and it is a useful screening test, mainly in regions of high disease prevalence [1], but this test is problematic with false positives, uncertain results and requiring electricity in the African field.[17] Consequently, better, more accurate serological tests have been developed recently.

Rapid Diagnostic Tests (RDTs) have now been developed and have considerable promise for HAT diagnostics. Following a prototype test called SD BIOLINE HAT which detects two trypanosome antigens (VSG LiTat 1.3 and VSG LiTat 1.5) [18] and which has a high specificity and sensitivity for HAT, a follow-up study[19] described a recombinant antigen based RDT that was even more sensitive than the SD BIOLINE HAT. This second-generation recombinant test is currently in use. However, it should be appreciated that both the CATT and these new RDTs may not necessarily detect a true active HAT infection, though they have a good negative predictive value, and also detect those patients who require treatment.[1]

Diagnostic staging of HAT is particularly difficult, and this is of considerable importance since treatment of severe late-stage disease is currently toxic. Many centres use the WHO definition of late-stage disease based on cerebrospinal fluid (CSF) analysis at lumbar puncture which is a CSF white blood cell (WBC) count of >5 per cu mm.[3] However, this criterion is problematic as not everyone accepts this definition and may use a different figure as high as 20 WBC per cu mm. There is also the complication of inter-observer variability in CSF cell counting. Unsurprisingly, there have been efforts to obtain a better definition of late-stage HAT, but if new tests are compared with the WHO definition, this itself comprises a form of circular argument. This problem has been addressed in detail [20] where it was suggested that novel statistical methodology might be used in correlating CSF WBC and other staging biomarkers. If there was available a non-toxic oral drug that was effective for both the milder and severe later forms of the encephalitic stage of HAT, then the need for a CSF analysis would be obviated. Although there have been described non-invasive methods to detect early and late-disease such as polysomnography [21] and, more recently, the use of a actigraphy sleep score,[22] while these are certainly useful adjuncts to diagnosis, it is very unlikely that they would be used to determine the use of late-stage treatment though they may well have a role in detecting disease relapses rather than repeat lumbar punctures.

NOVEL THERAPIES FOR HAT

Treatment of the early-stage of HAT consists of suramin in the case of *T.b.rhdesiense* and pentamidine for *T.b.gambiense* .[2] While neither of these drugs are completely free of toxic effects, they are widely used and are not generally regarded as problematic. The treatment of late-stage disease, however, is more difficult and controversial, and has been the subject of much recent research. For many years since 1949 the arsenical drug melarsorol was given systemically for the encephalitic stage of both *rhodesiense* and *gambiense* HAT. While very effective at treating the disease, it is unfortunately very painful to administer, and has several side effects the most serious of which is a severe post-treatment reactive encephalopathy (PTRE) which occurs in about 10% patients, half of whom will die.[1] Much recent effort, therefore, has gone into developing novel and les toxic therapies for late-stage disease.

An important advance was the development of treatment of *T.b.gambiense* with NECT (nifurtimox-eflornithine combination therapy) which is as effective as melaroprol, and less toxic.[23] While ineffective against *T.b.rhodesiense*, and requiring 14 days of systemic treatment with eflornithine as well as not being completely free of toxic effects, NECT has been for many years first line therapy for late-stage *gambiense* HAT.[1] A more recent advance in this field was the use of oral fexinidazole for *gambiense* disease. In a multicentre, randomised, phase 2/3, open-label, non-inferiority trial in the DRC, oral fexinidazole was reported to be effective and safe for the treatment of *T b gambiense* infection compared with NECT in late-stage HAT patients.[24] However, 9% of patients receiving fexinidazole were treatment failures, and the efficacy of fexinidazole was less than that seen in NECT. Such treatment failures should be given NECT. An oral drug clearly has advantages over systemically administered drugs, and at present it seems reasonable to treat *gambiense* HAT which is in the milder, early part of the encephalitic stage (with a CSF WBC of <100 per cu mm) with fexinidazole.[25] More severe cases (with a CSF WBC of >100 per cu mm) should instead be treated with NECT as first line therapy. A follow-up study reported in a multicentre, single-arm, open-label, phase 2–3 trial showed that oral fexinidazole was a safe and effective first-line treatment option across all *gambiense* HAT disease stages in paediatric patients.[26] Whether current trials of the use of oral fexinidazole in *T.b rhodesiense* HAT will be as effective as in *gambiense* disease remains to be determined. Fexinidazole should only be administered under supervision of trained health staff.[25] At present, melarsoprol remans the first line therapy for late- stage *rhodesiense* infection., but this may change in the future.

Another very promising drug candidate is a compound of the oxaborole group. In a recently published report, in a a multicentre, open-label, single-arm, phase 2/3 trial, both the efficacy and safety of acoziborole was evaluated in patients with HAT caused by *T.b.gambiense*.[27] It was found that there was a high efficacy and favourable safety profile of the treatment so that acoziborole holds promise in the management of early and late-stage HAT. However, a total of 38 drug-related treatment-emergent adverse events occurred in 29 (14%) patients, though these were all mild or moderate. A different therapeutic approach was to develop cyclodextrin-melarsoprol inclusion complexes, which are oral drugs that retain the high efficacy of melarsoprol while losing its high toxicity when tested in an experimental mouse model of HAT.[28] Oral complexed melarsoprol has been given dual status by the European Medical Agency (EMA) and the US Food and Drug Administration (FDA) as an orphan drug for treating African sleeping sickness, However, despite its considerable promise of oral complexed melarsoprol, clinical trials in Africa have stalled due to the unwillingness of research councils and pharmaceutical forms to fund them.

NEUROPATHOGENESIS

Advances in understanding the neuropathogenesis of HAT have been steady but incremental over the last decade. Most of our understanding has come from experimental animal studies in which mechanistic experiments can be carried out, rather than simple analyses of blood, urine and CSF obtained from infected patients. The role of different cytokines is prominent with a balance of counter and pro-inflammatory cytokines being crucial to the generation of neuroinflammation and disease outcome. This is shown diagrammatically in Figure 1, and the relative importance of Substance P[29] and kynurenine pathway metabolites in late-stage disease pathogenesis[30] remains to be determined.

Two aspects of neuropathogenesis have become particularly important over the last decade, namely the role of the BBB in which the trypanosomes cross into the CNS, and the importance of innate immunity. The timing and mechanism of parasite traversal through a damaged BBB is crucial to our understanding of encephalitic stage disease. However, this process must be quite complex as *in vitro* models have shown that the trypanosomes are able to cross the microvascular endothelial cell layer of the human brain producing only a brief impairment of BB integrity.[31] New imaging tools have been applied to address the issue of BBB integrity. Magnetic Resonance imaging (MRI) of infected humans is not feasible in the African field but is used in Western medical centres, and in experimental mouse models where it has been shown that leakage of the BBB occurs prior to CNS disease .[32] MRI has recently been used to image an individual organ, the spleen, in experimental trypanosomiasis.[33] Also, multi-photon microscopy and bioluminescent labelled trypanosomes have been used to track parasite invasion and to develop methods of assessing the efficacy of new drugs.[34] There is now mounting evidence from previous studies [34,35] that trypanosomes enter the CNS earlier than previously thought, which is consistent with the detection of neurological features in early-stage disease.[12] Further, a recent study using gene microarray profiling, showed that both neuronal genes and immune response genes were activated early during infection and prior to CNS disease, thereby impugning the whole notion of early and late-stage disease.[37]

Another relatively recent finding is the presence of trypanosomes in the skin of aparasitaemic and undiagnosed individuals.[37] This has been an overlooked reservoir of human infection the significance of which is potentially very significant. Further, these authors also showed that after experimental mouse infection, large numbers of trypanosomes were detectable in the skin, and that these could be transmitted to the tsetse fly vector.

Genetically modified trypanosomes and experimental hosts have emphasised the importance of innate, as well as adaptive, immune mechanisms in trypanosomiasis pathogenesis. Using these kinds of techniques, it has been shown, for instance, that experimentally infected mice which were genetically engineered to 'knockout' the innate signalling molecule Toll-like receptor- 9, showed a reduced number of T lymphocytes compared to wild-type mice.[38] Further, knocking out the important MyD88 adaptor protein in infected mice resulted in a reduction in T cells but an increase in the number of parasites in the brain compared to controls.[38,39] Over the next few years, it is likely that the precise role of the various innate immune mechanisms, together with their interaction with other components of the immune response, will be clarified further during experimental trypanosome infection.

PROSPECTS FOR THE FUTURE AND CONCLUSIONS

The WHO has two main goals for HAT infection. First is the elimination of HAT as a public health problem by 2020. This goal has largely been accomplished although several African countries have yet to submit their incidence figures to WHO for final validation. The second goal is to achieve interruption of disease transmission by 2030. While this is a possibility, it is very ambitious and may well require a change in general strategy. There are several difficulties that arise in this context. First, the animal reservoirs for *T.b.rhodesiense* are cattle so it will be very difficult to eradicate infected animals by this time, something that will also require much better drug treatment of animal trypanosomiasis. The recent observation that some humans have reservoirs of trypanosomes in their skin is also a significant issue in terms of parasite eradication. Second, there may be underdiagnosis and under-detection of new cases particularly as the early-stage symptoms may be non-specific. Third, the eradication of the tsetse fly vector may prove to be very difficult as the area infested is so large. Fourth, HAT has already demonstrated its ability to wax and wane and cause intermittent epidemics and resurgences so that very careful monitoring of individuals in sub-Saharan Africa will be required long term. This second goal may yet be achieved for *gambiense* disease at least, but only time will tell whether this ambitious second aim can be attained

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Conflicts of Interest I declare that I have no conflicts of interest.
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LEGEND FOR FIGURE

Schematic representation of immunological interactions in Trypanosomiasis pathogenesis. VSG: variable surface glycoprotein. NK: natural Killer. tltf : trypanosome-derived lymphocyte triggering factor.(From reference 3).

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